palimpsestr: User Manual

Complete Reference Guide v0.18.0

Enzo Cocca

2026-06-14

Contents

1 Introduction

palimpsestr is an R package for the probabilistic decomposition of archaeological palimpsests using the Stratigraphic Entanglement Field (SEF) framework. It integrates spatial proximity, vertical distribution, chronological overlap, and cultural similarity to estimate latent depositional phases.

1.1 Installation

```r
# From GitHub
devtools::install_github("enzococca/palimpsestr")

# From local source
install.packages("palimpsestr_0.18.0.tar.gz", repos = NULL, type = "source")
```

1.2 Required and optional packages

Package	Status	Purpose
stats, utils, graphics, grDevices	Required	Core R
ggplot2	Suggested	Publication-quality plots (`gg_*` functions)
plotly	Suggested	Interactive visualization (`as_plotly()`)
sf	Suggested	GIS export (`as_sf_phase()`, `as_sf_links()`)
DBI, RSQLite, RPostgres	Suggested	Database import (`read_db()`)

2 Data Preparation

2.1 Required columns

Every dataset must contain:

Column	Type	Description
`x`	numeric	Easting coordinate (metres)
`y`	numeric	Northing coordinate (metres)
`z`	numeric	Depth or elevation (metres)
`date_min`	numeric	Start of chronological interval
`date_max`	numeric	End of chronological interval
`class`	character	Material class (ceramic, lithic, bone, metal, etc.)

2.2 Optional columns

Column	Type	Description
`id`	any	Unique identifier for each find
`context`	character	Stratigraphic unit label (e.g., "SU_1")
`taf_score`	numeric (0–1)	Taphonomic disturbance score
`true_phase`	integer	Known phase (for validation only)

2.3 Notes on Z coordinates

Each artifact needs its own z value. If you don't have point-level depth measurements, you can assign the mean depth of the stratigraphic unit (US) to all finds within it:

```
# Example: assign US mean depth to finds
data$z <- ave(data$z_us, data$context, FUN = mean)
```

2.4 Calibrated dates: rcarbon and OxCal

Calibrated radiocarbon or OxCal-modelled dates can be converted to the `date_min`/`date_max`/`date_mid` columns expected by `fit_sef()`:

```
# From rcarbon
cal <- rcarbon::calibrate(x = c(2500, 2400), errors = c(30, 30))
chronology_from_rcarbon(cal)

# From OxCal (an oxcAAR result, or a data.frame of start/end ranges)
ranges <- data.frame(id = c("a", "b"), start = c(-700, -50), end = c(-600, 100))
chronology_from_oxcal(ranges)
```

2.5 Simulating test data

```r
# Well-separated phases (low mixing)
easy <- archaeo_sim(n = 150, k = 3, seed = 1, mixing = 0.05)

# Moderate palimpsest
moderate <- archaeo_sim(n = 200, k = 3, seed = 1, mixing = 0.30)

# Compressed palimpsest (hard to resolve)
compressed <- archaeo_sim(n = 250, k = 4, seed = 1, mixing = 0.50)
```

Parameters:

- `n`: number of artifacts
- `k`: number of phases
- `seed`: random seed for reproducibility
- `mixing`: proportion of inter-phase mixing (0 = none, 1 = complete)

2.6 Built-in datasets

The package includes four built-in datasets:

Dataset	Finds	Phases	Mixing	Description
`demo_easy`	150	3	5%	Well-separated phases
`demo_moderate`	200	3	30%	Moderate palimpsest
`demo_compressed`	250	4	50%	Heavily disturbed
`villa_romana`	615	–	Real data	Poggio Gramignano Roman villa (VRPG 3004)

```r
data(villa_romana)
str(villa_romana)
#> 'data.frame':	615 obs. of  9 variables:
#>  $ id       : chr  "VRPG_0001" "VRPG_0002" "VRPG_0003" "VRPG_0004" ...
#>  $ x        : num  2299041 2299041 2299041 2299041 2299041 ...
#>  $ y        : num  4714126 4714126 4714126 4714126 4714126 ...
#>  $ z        : num  284 284 284 284 284 ...
#>  $ context  : chr  "US_102" "US_102" "US_102" "US_102" ...
#>  $ date_min : num  425 425 425 425 425 425 425 425 425 425 ...
#>  $ date_max : num  450 450 450 450 450 450 450 450 450 450 ...
#>  $ class    : chr  "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" "Reperto Ceramico" ...
#>  $ taf_score: num  0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 0.7 ...
table(villa_romana$class)
#> 
#>              Campionature           Intonaco Dipinto Materiale Da Costruzione 
#>                         5                          1                       38 
#>         Materiale Lapideo         Pietre Dure-Gemme    Reperti Antropologici 
#>                         1                          1                        1 
#>  Reperti Archeozoologici          Reperto Anforaceo         Reperto Ceramico 
#>                         7                         76                      417 
#>       Reperto Faunistico            Reperto Lapideo        Reperto Metallico 
#>                         1                          6                       19 
#>      Reperto Numismatico             Reperto Vitreo  Resti Di Pavimentazioni 
#>                        10                         25                        1
```

```
#>     Scarti Di Produzione                   Stucco
#>                        5                        1
# Core Analysis

## Fitting the SEF model: `fit_sef()`
```

```r
fit <- fit_sef(
  data      = moderate,
  coords    = c("x", "y", "z"),       # coordinate columns
  chrono    = c("date_min", "date_max"), # chronological range columns
  class     = "class",                  # material class column
  tafonomy = "taf_score",              # optional: taphonomic score
  context   = "context",               # optional: stratigraphic unit
  harris    = NULL,                     # optional: Harris Matrix penalty
  k         = 3,                        # number of phases
  weights   = c(ws = 1, wz = 1, wt = 1, wc = 1), # domain weights
  seed      = 42,                       # random seed
  em_iter   = 100,                      # max EM iterations
  em_tol    = 1e-5,                     # convergence tolerance
  n_init    = 5                         # number of random initialisations
)
print(fit)
#> <sef_fit>
#> Observations: 200
#> Estimated phases: 3
#> Mean entropy: 0.000
#> Mean local SEI: 249.857
#> Mean energy: 2.008
#> LogLik: -1177.707
#> BIC: 2572.646
#> ICL: 2572.643
#> PDI: 1.000
#> Converged: yes
#> Class model: multinomial
#> Initialisations: 5
```

2.6.1 Parameters explained

Parameter	Default	Description
`k`	3	Number of depositional phases to estimate
`weights`	`c(ws=1, wz=1, wt=1, wc=1)`	Weights for spatial (`ws`), vertical (`wz`), temporal (`wt`), and cultural (`wc`) evidence. Since v0.14 they scale the feature dimensions and enter the likelihood, so they can be cross-validated with `optimize_weights()`
`em_iter`	100	Maximum EM iterations
`em_tol`	1e-5	Convergence tolerance (log-likelihood change)
`n_init`	5	Number of random initialisations (higher = more robust, slower)

Parameter	Default	Description
`tafonomy`	NULL	Column name for taphonomic disturbance (0–1)
`context`	NULL	Column name for stratigraphic unit labels
`harris`	NULL	$n \times n$ penalty matrix (from `harris_from_contexts()` or `read_harris()`)

2.6.2 Statistical-model options

These options (all optional, sensible defaults) control the statistical core:

Parameter	Default	Description
`class_model`	`"multinomial"`	How the material class is modelled. `"multinomial"` (recommended) treats it as a per-phase categorical distribution, so typology does not fragment stratigraphic units; `"gaussian"` is the legacy one-hot behaviour
`class_smoothing`	1	Dirichlet pseudo-count for the per-phase class probabilities (shrunk towards the global class frequencies)
`var_structure`	`"diagonal"`	`"diagonal"` (free per-dimension variances) or `"spherical"` (one shared variance per phase; the structure under which `weights` are identifiable)
`chrono_uncertainty`	FALSE	If TRUE, propagate each find's dating-interval width $(t_{max} - t_{min})^2/12$ into the temporal likelihood, so coarsely dated finds rely less on chronology
`strat_dynamic`	FALSE	If TRUE (with `context`), apply the stratigraphic constraint as a Neighborhood-EM field recomputed each E-step instead of a static penalty frozen at initialisation
`strat_beta`	1	Strength of the dynamic stratigraphic field
`noise`	FALSE	If TRUE, add a uniform background component (Fraley and Raftery, 1998); `$noise_prob` becomes a genuine outlier probability and extreme finds are kept out of the phase estimates
`noise_prior`	0.05	Initial mixing weight of the noise component
`chrono_precision`	FALSE	Add $1/t_{span}$ as a feature (precise dates weighted higher)
`taf_as_feature`	FALSE	Add `taf_score` as an explicit feature dimension
`residuality`	FALSE	Add a per-find date-vs-context-mean mismatch feature (needs `context`)

Parameter	Default	Description
`class_scale`	FALSE	Scale one-hot class columns by $1/\sqrt{n_{classes}}$ (only relevant with `class_model = "gaussian"`)
`subclass`	NULL	Optional sub-class column used instead of `class` for the categorical/one-hot block

`recommend_setup(data, context = "context", tafonomy = "taf_score")` inspects a dataset and suggests which of these to enable, flagging caveats such as unit-level coordinates or unit-tied chronology.

2.6.3 Output: the `sef_fit` object

The returned object contains:

Field	Type	Description
`$phase`	integer vector	Dominant phase assignment per find
`$phase_prob`	matrix	$n \times k$ matrix of phase membership probabilities (conditional on not being noise when `noise = TRUE`)
`$cat_prob`	matrix	$k \times L$ per-phase class profile (multinomial model); see `phase_composition()`
`$noise_prob`	numeric vector	Per-find outlier posterior (when `noise = TRUE`)
`$entropy`	numeric vector	Shannon entropy per find
`$energy`	numeric vector	ESE per find
`$local_sei`	numeric vector	Local SEI per find
`$converged`	logical	Whether EM converged
`$n_init`	integer	Number of initialisations used
`$em_loglik`	numeric vector	Log-likelihood trace across iterations
`$model_stats`	list	Summary statistics (PDI, BIC, ICL, loglik, etc.)

The per-phase class composition can be read directly:

```
# Per-phase typological profile (multinomial model)
phase_composition(fit)
#>   phase      bone   ceramic    lithic     metal
#> 1     1 0.2429904 0.3573578 0.2118628 0.1877889
#> 2     2 0.1578399 0.3261697 0.3050742 0.2109162
#> 3     3 0.2041745 0.3028718 0.3629811 0.1299727
```

2.7 Phase count selection: `compare_k()`

When K is unknown, compare multiple values:

```
ck <- compare_k(moderate, k_values = 2:5, seed = 42,
                tafonomy = "taf_score", context = "context")
print(ck)
#>   k       pdi mean_entropy mean_local_sei mean_energy    loglik      bic
#> 1 2 0.9929875 4.860672e-03       249.8574    2.008381 -1426.116 2995.288
#> 2 3 0.9999945 6.095726e-06       249.8574    2.008381 -1177.707 2572.646
#> 3 4 0.9438440 7.784876e-02       249.8574    2.008381 -1156.909 2605.225
#> 4 5 0.9160600 1.350962e-01       249.8574    2.008381 -1114.002 2593.588
```

```
#>        icl tot_withinss pseudo_bic
#> 1 2993.343     603.9637   274.0225
#> 2 2572.643     400.1314   218.1699
#> 3 2574.085     341.0552   212.7117
#> 4 2539.550     292.7963   208.6899
```

The optimal K minimises BIC while maximising PDI.

2.8 Key diagnostics

```
# Palimpsest Dissolution Index (global separability, 0--1)
pdi(fit)
#> [1] 0.9999945

# Compact summary
sef_summary(fit)
#> $n
#> [1] 200
#>
#> $k
#> [1] 3
#>
#> $pdi
#> [1] 0.9999945
#>
#> $mean_entropy
#> [1] 6.095726e-06
#>
#> $mean_energy
#> [1] 2.008381
#>
#> $loglik
#> [1] -1177.707
#>
#> $bic
#> [1] 2572.646
#>
#> $phase_count
#>
#>  1  2  3
#> 67 66 67

# Phase probability table
head(as_phase_table(fit))
#>     id dominant_phase       phase1       phase2 phase3      entropy local_sei
#> 1 F001              3 1.124319e-19 2.076556e-40      1 4.905622e-18  252.1957
#> 2 F002              3 7.713722e-18 4.721978e-40      1 3.039479e-16  248.5608
#> 3 F003              3 4.164452e-21 2.368336e-44      1 1.954282e-19  239.7297
#> 4 F004              3 2.645145e-23 1.267429e-46      1 1.375124e-21  260.5879
#> 5 F005              3 2.204926e-15 5.781533e-35      1 7.663248e-14  279.3199
#> 6 F006              3 1.911759e-18 7.612349e-42      1 7.799693e-17  284.5397
#>     energy
#> 1 2.403297
```

```
#> 2 1.756757
#> 3 1.931740
#> 4 1.854746
#> 5 2.362912
#> 6 2.002368

# Full diagnostic table (input + diagnostics)
head(phase_diagnostic_table(fit))
#>     id        x         y        z context date_min date_max   class taf_score
#> 1 F001 12.36888 104.53174 11.79768    SU_4 970.6746 1011.055  lithic 0.4687699
#> 2 F002 19.12233  89.86743 19.78304    SU_3 960.5955 1080.232  lithic 0.3796706
#> 3 F003 24.19310  96.13386 18.97572    SU_7 966.4167 1030.552    bone 0.2288079
#> 4 F004 26.50473  99.62853 18.50500    SU_2 947.1761 1027.755  lithic 0.2246840
#> 5 F005 45.78809  91.97095 17.58970    SU_1 975.1793 1071.124 ceramic 0.1925518
#> 6 F006 32.65961  89.87875 17.67439    SU_4 946.7372 1055.110 ceramic 0.5558153
#>   true_phase dominant_phase       phase1       phase2 phase3      entropy
#> 1          1              3 1.124319e-19 2.076556e-40      1 4.905622e-18
#> 2          1              3 7.713722e-18 4.721978e-40      1 3.039479e-16
#> 3          1              3 4.164452e-21 2.368336e-44      1 1.954282e-19
#> 4          1              3 2.645145e-23 1.267429e-46      1 1.375124e-21
#> 5          1              3 2.204926e-15 5.781533e-35      1 7.663248e-14
#> 6          1              3 1.911759e-18 7.612349e-42      1 7.799693e-17
#>   local_sei   energy
#> 1  252.1957 2.403297
#> 2  248.5608 1.756757
#> 3  239.7297 1.931740
#> 4  260.5879 1.854746
#> 5  279.3199 2.362912
#> 6  284.5397 2.002368
```

3 Intrusion Detection

`detect_intrusions()` returns, per find, an `intrusion_prob`, a `direction` factor (`older_than_context` / `in_context` / `younger_than_context`), and a signed `chrono_gap` (years). When the model was fitted with `noise = TRUE`, `intrusion_prob` is the noise-component posterior — a genuine probability of being an outlier; otherwise it is a heuristic composite of entropy, energy, and inverse local SEI.

```
fit_n <- fit_sef(moderate, k = 3, tafonomy = "taf_score",
                 context = "context", noise = TRUE, seed = 42)
di <- detect_intrusions(fit_n)
suspects <- di[di$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects), "\n")
#> Suspected intrusions: 24
head(suspects[order(suspects$intrusion_prob, decreasing = TRUE), ])
#>       id intrusion_prob  direction chrono_gap     intrusion_type
#> 13  F013      1.0000000 in_context          0 outlier_in_context
#> 24  F024      0.9999992 in_context          0 outlier_in_context
#> 112 F112      0.9999973 in_context          0 outlier_in_context
#> 1   F001      0.9999948 in_context          0 outlier_in_context
#> 145 F145      0.9999412 in_context          0 outlier_in_context
#> 44  F044      0.9998334 in_context          0 outlier_in_context
```

The `direction` distinguishes residual material (`older_than_context`) from possible latent features (`younger_than_context`); it is informative only when finds within a unit carry distinct chronologies (not under unit-tied dating).

Since v0.19, `detect_intrusions()` also returns an `intrusion_type` factor that couples the intrusion magnitude with the direction: `residual` (flagged and older-than-context), `latent_feature` (flagged and younger-than-context), `outlier_in_context` (flagged but undirected), or `not_flagged`. `gg_outliers()` colours finds by this type.

4 Model Validation

When true phase labels are known:

```r
# Adjusted Rand Index (-1 to 1, 1 = perfect)
adjusted_rand_index(fit, moderate$true_phase)
#> [1] 1

# Confusion matrix (auto-reordered for best diagonal match)
confusion_matrix(fit, moderate$true_phase)
#>          true
#> estimated  2  3  1
#>         1 67  0  0
#>         2  0 66  0
#>         3  0  0 67
```

5 Structured Exports

5.1 Summary per stratigraphic unit

```r
us_summary_table(fit)
#>   context n_finds dominant_phase    purity mean_entropy mean_energy
#> 1    SU_1      18              3 0.4444444 6.763679e-05    1.933266
#> 2    SU_2      29              2 0.3793103 2.610162e-12    2.009269
#> 3    SU_3      20              2 0.3500000 2.617318e-11    2.160271
#> 4    SU_4      20              1 0.4500000 3.112000e-09    2.083647
#> 5    SU_5      16              1 0.3750000 2.177784e-11    2.005937
#> 6    SU_6      29              1 0.3793103 4.345181e-08    1.988188
#> 7    SU_7      23              2 0.4347826 1.735165e-09    2.039224
#> 8    SU_8      16              1 0.5625000 3.867276e-10    1.920946
#> 9    SU_9      29              3 0.4137931 1.081262e-08    1.942775
#>   mean_local_sei n_intrusions
#> 1       250.2169            3
#> 2       245.1726            3
#> 3       244.9395            2
#> 4       250.8428            0
#> 5       255.8361            0
#> 6       253.5263            0
#> 7       256.1566            0
#> 8       257.4712            0
#> 9       240.8673            3
```

Columns: `context`, `n_finds`, `dominant_phase`, `purity`, `mean_entropy`, `mean_energy`, `mean_local_sei`, `n_intrusions`.

5.2 Phase transition matrix

Shows how often finds from phase i sit directly above finds from phase j:

```r
phase_transition_matrix(fit)
#>        phase1 phase2 phase3
#> phase1     28     35      3
#> phase2     33     31      2
#> phase3      6      0     61
```

5.3 Export all results to CSV

```r
export_results(fit, dir = "results/", format = "csv", prefix = "site_X")
```

Creates 4 files:

- `site_X_phases.csv` — phase assignments and diagnostics per find
- `site_X_intrusions.csv` — intrusion probabilities
- `site_X_us_summary.csv` — per-US aggregated statistics
- `site_X_model_summary.csv` — model-level metrics (n, k, PDI, BIC, etc.)

6 Harris Matrix Integration

6.1 Auto-generate from depth ordering

```r
H <- harris_from_contexts(moderate, z_col = "z", context_col = "context",
                          penalty_scale = 0.5)
dim(H)
#> [1] 200 200
```

6.2 Import from CSV

The CSV must have columns `from`, `to`, and optionally `weight`:

```
from,to,weight
SU_1,SU_2,1
SU_2,SU_3,1
SU_1,SU_3,0.5
```

```r
H <- read_harris("harris_matrix.csv", contexts = data$context)
fit <- fit_sef(data, k = 3, harris = H, context = "context")
```

6.3 Validate phases against stratigraphy

```r
validate_phases_harris(fit)
#>   upper lower upper_phase lower_phase consistent
#> 1  SU_1  SU_2           3           2      FALSE
#> 2  SU_2  SU_5           2           1      FALSE
#> 3  SU_5  SU_7           1           2       TRUE
#> 4  SU_7  SU_6           2           1      FALSE
#> 5  SU_6  SU_4           1           1       TRUE
#> 6  SU_4  SU_9           1           3       TRUE
#> 7  SU_9  SU_3           3           2      FALSE
#> 8  SU_3  SU_8           2           1      FALSE
```

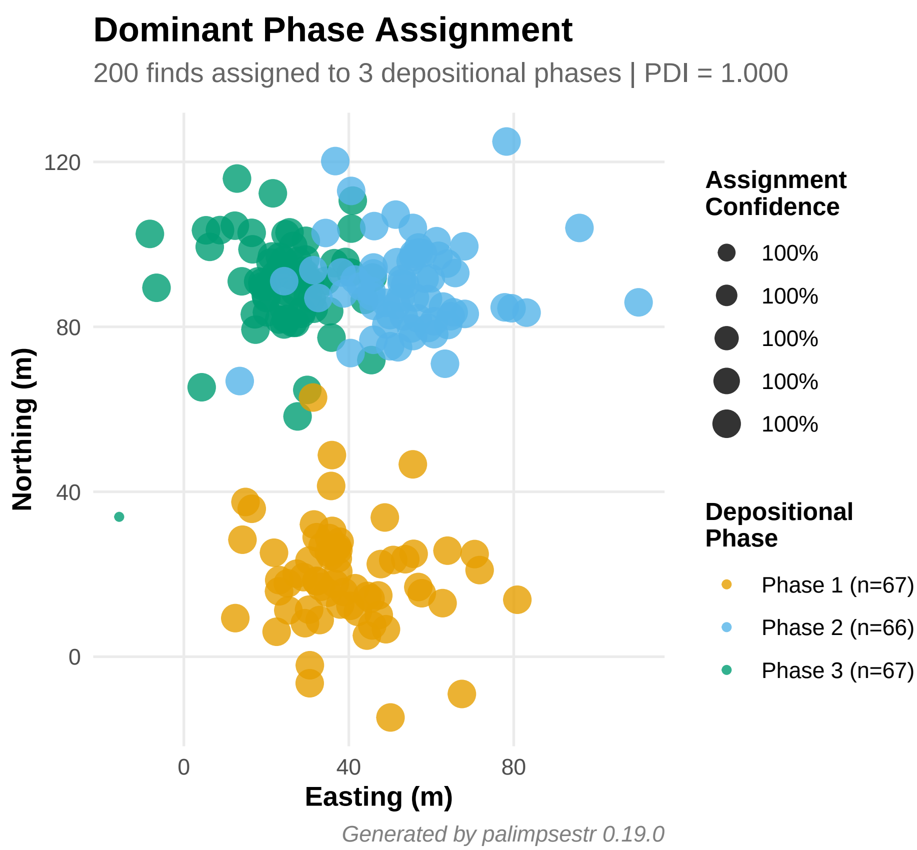
The `consistent` column flags whether the dominant phase ordering respects stratigraphic depth. `FALSE` indicates an inversion that may warrant investigation.

7 Visualization

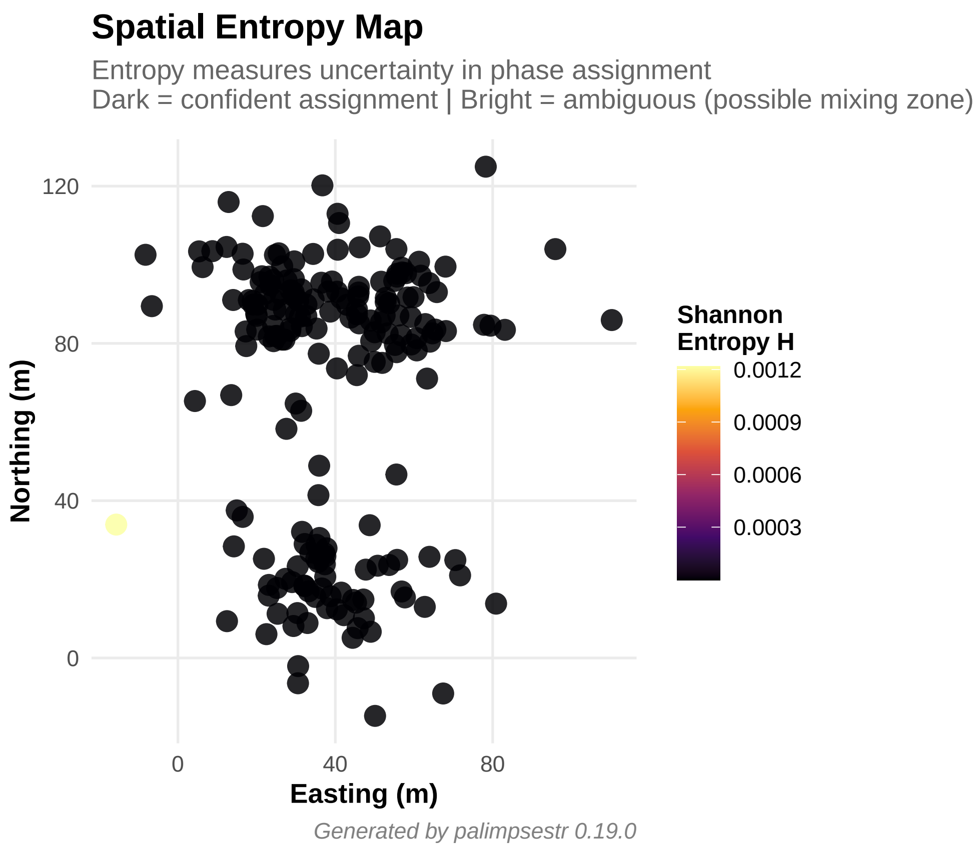
7.1 ggplot2 plots (static)

All `gg_*` functions require **ggplot2** and return ggplot objects.

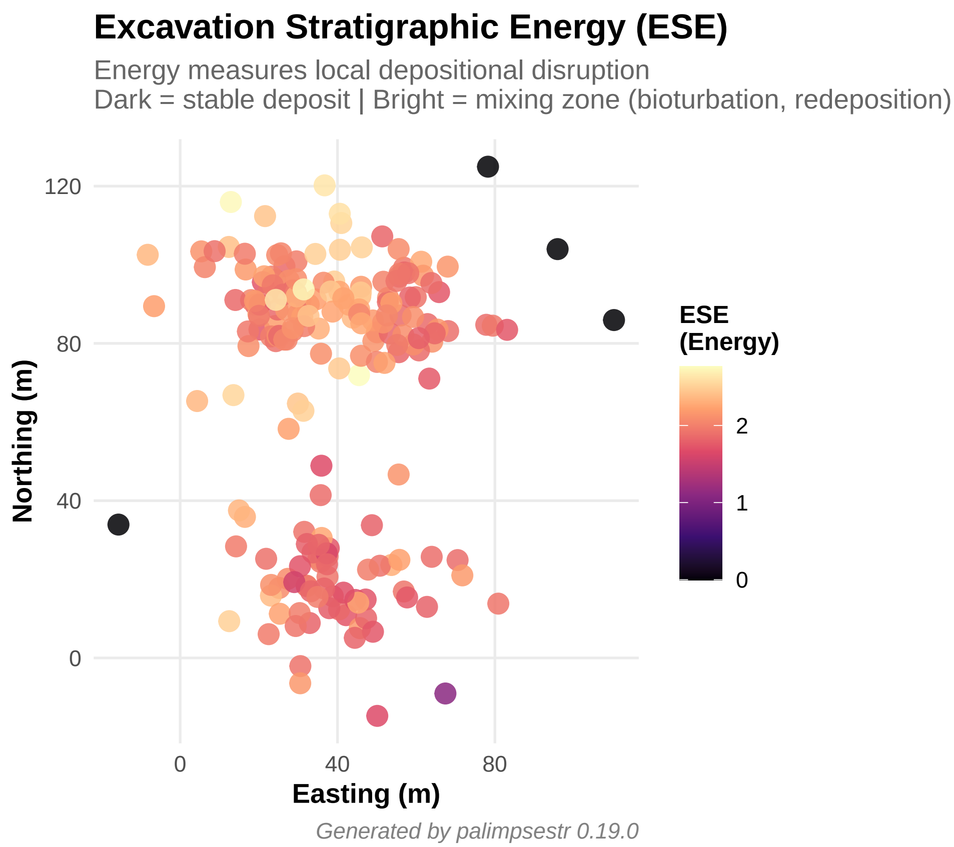
```r
library(ggplot2)

# Phase assignment map
gg_phasefield(fit)
```



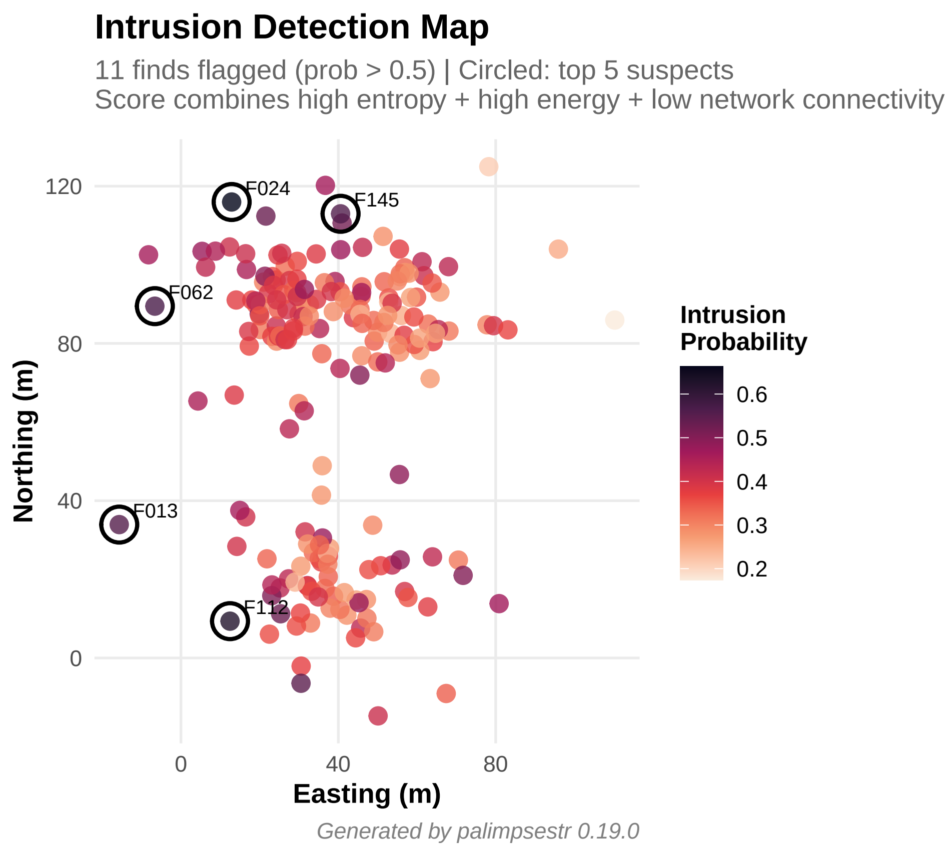

```r
# Entropy map (uncertainty)
gg_entropy(fit)
```



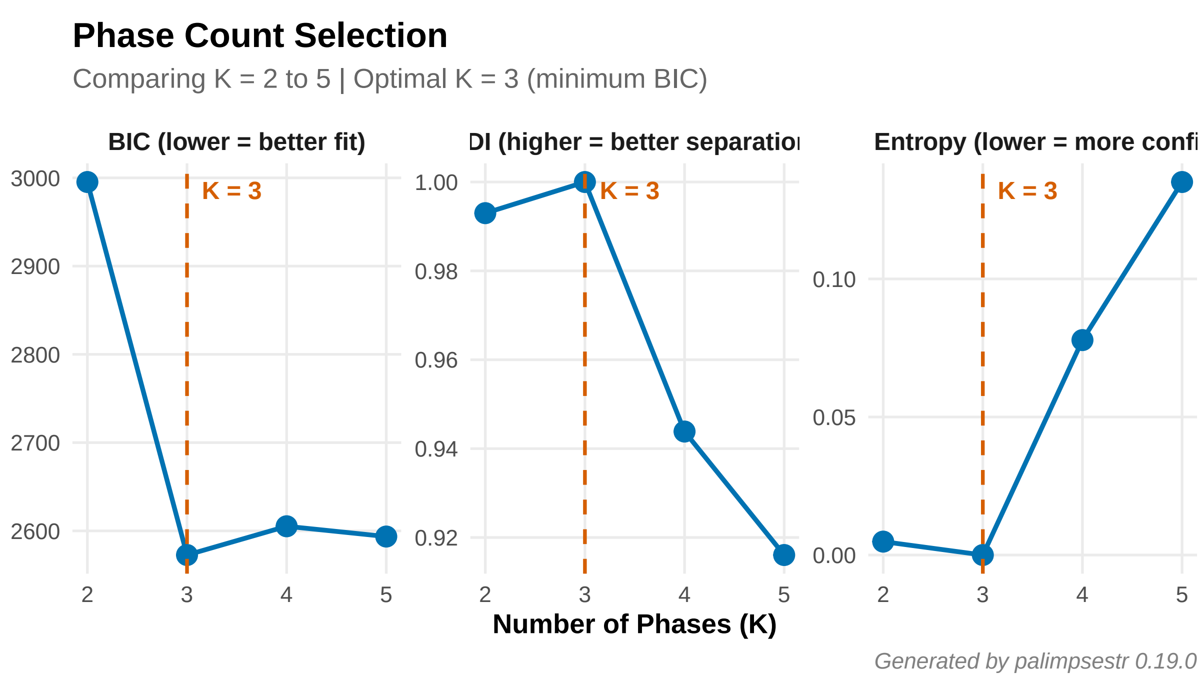

```r
# Energy map (local disruption)
gg_energy(fit)
```



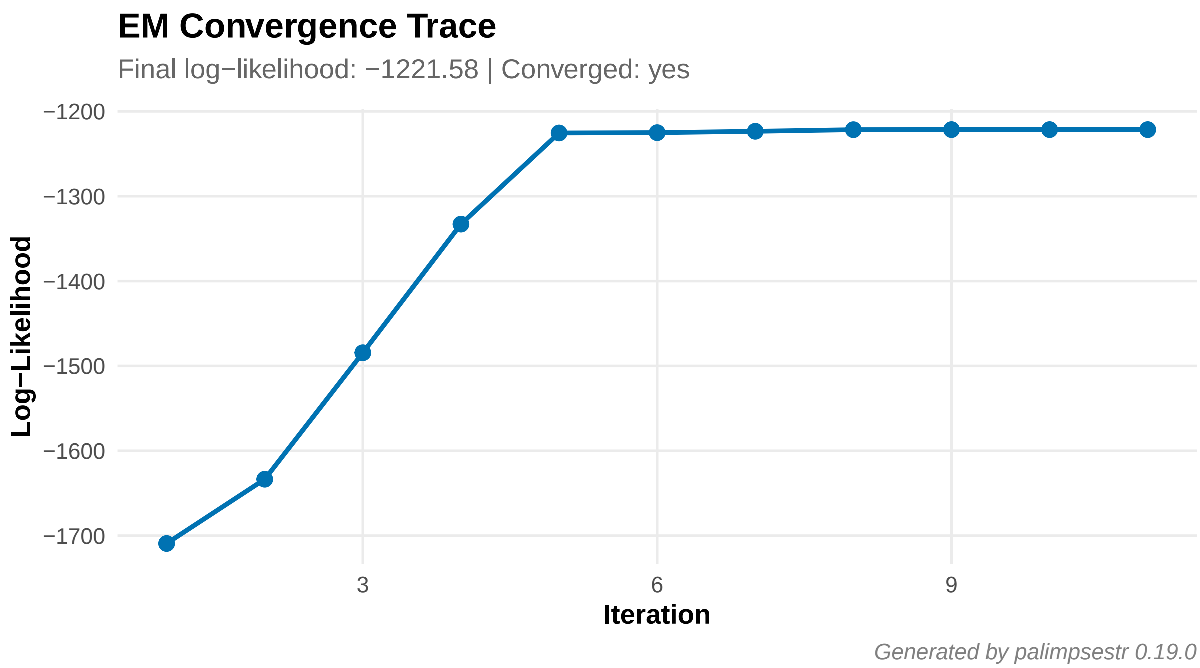

```r
# Intrusion detection map
gg_intrusions(fit, top_n = 5)
```



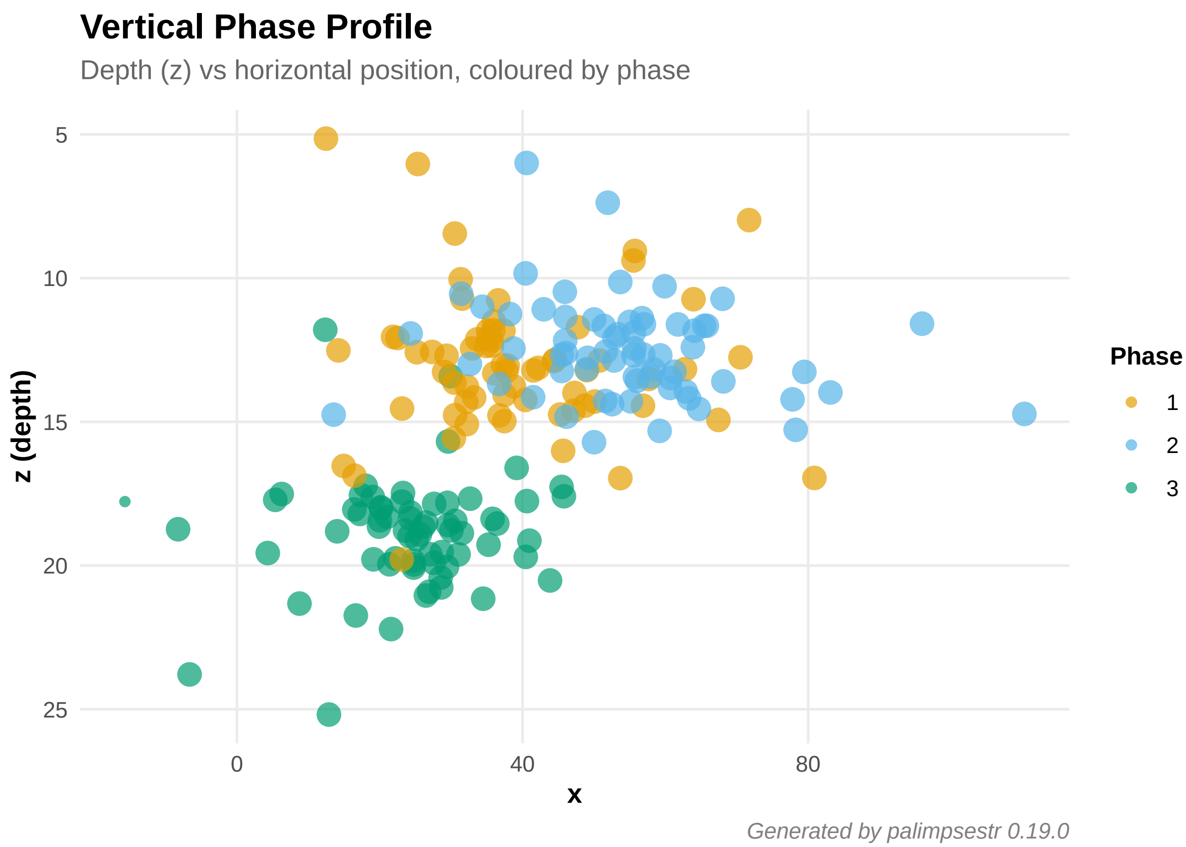

```r
# Phase count selection diagnostics
gg_compare_k(ck)
```



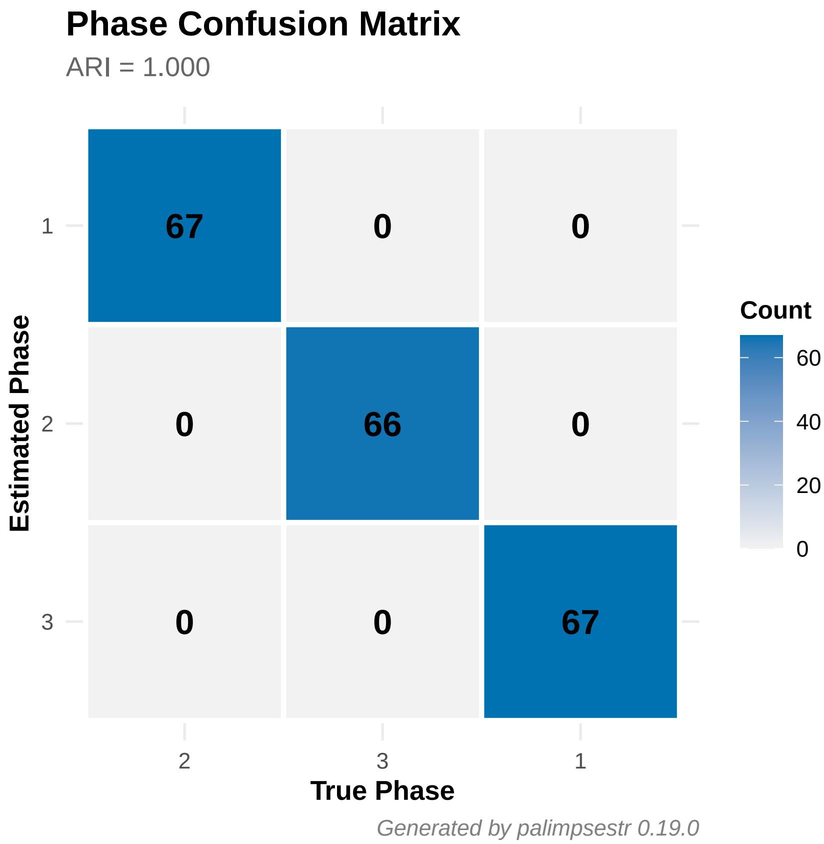
```r
# EM convergence trace
gg_convergence(fit)
```




```r
# Vertical phase profile (depth vs horizontal)
gg_phase_profile(fit)
```




```r
# Confusion matrix heatmap
gg_confusion(fit, moderate$true_phase)
```



7.1.1 Model-output diagnostic plots (v0.18)

Four plots visualise the outputs of the refined statistical model:

```r
# Per-phase class composition (multinomial cat_prob), heatmap by default
gg_phase_composition(fit)
```


```r
# Within-unit phase coherence; pass compare = a gaussian-class fit to contrast
gg_unit_coherence(fit)

# Non-spatial ranking of the intrusion probability (noise posterior if noise=TRUE)
gg_outliers(fit_n)

# Directional intrusions: residual vs latent (needs per-find dating)
gg_direction(fit_n)
```

7.1.2 Type longevity

`type_longevity()` aggregates the posterior to estimate the temporal envelope of each material class; `gg_longevity()` draws it as a Gantt-style chart:

```r
tl <- type_longevity(fit)
gg_longevity(tl)
```

7.2 Interactive plots (plotly)

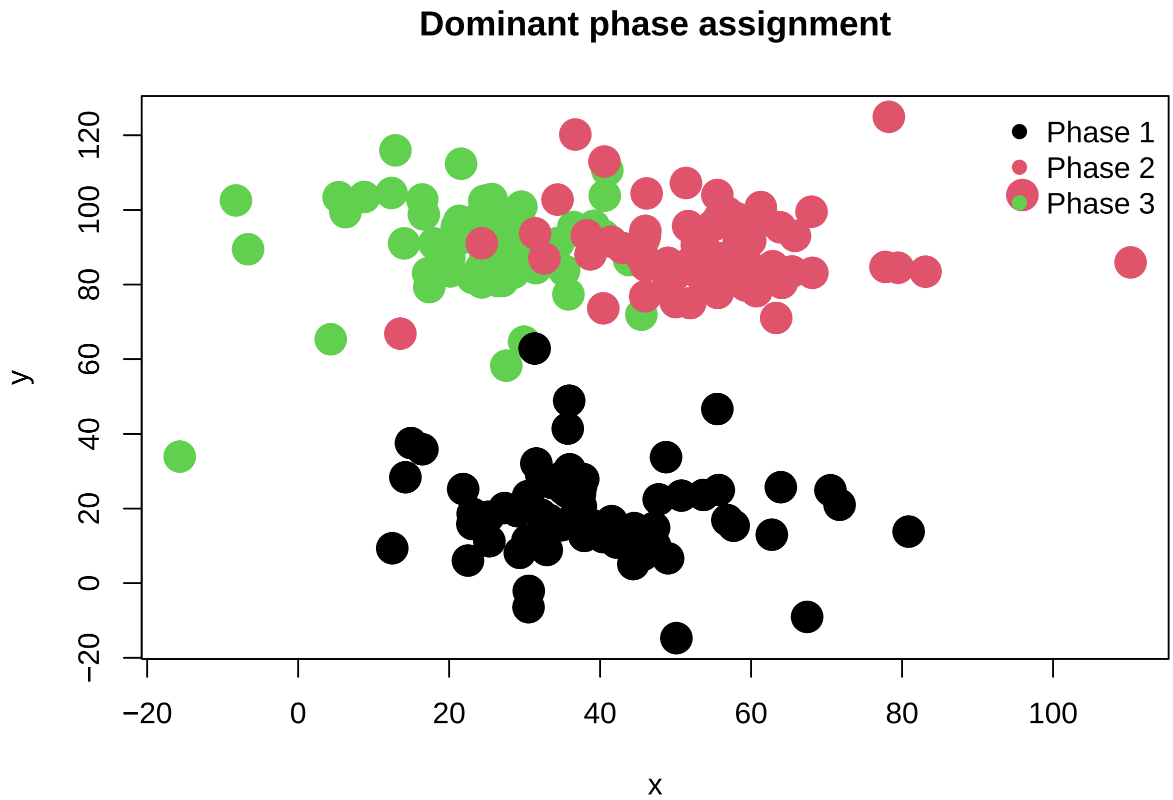
```r
library(plotly)

# Convert any gg_* plot to interactive
as_plotly(gg_phasefield(fit))
as_plotly(gg_entropy(fit))
as_plotly(gg_intrusions(fit))
```

Hover tooltips show: ID, context, phase probability, dating, class, entropy, energy, intrusion score.

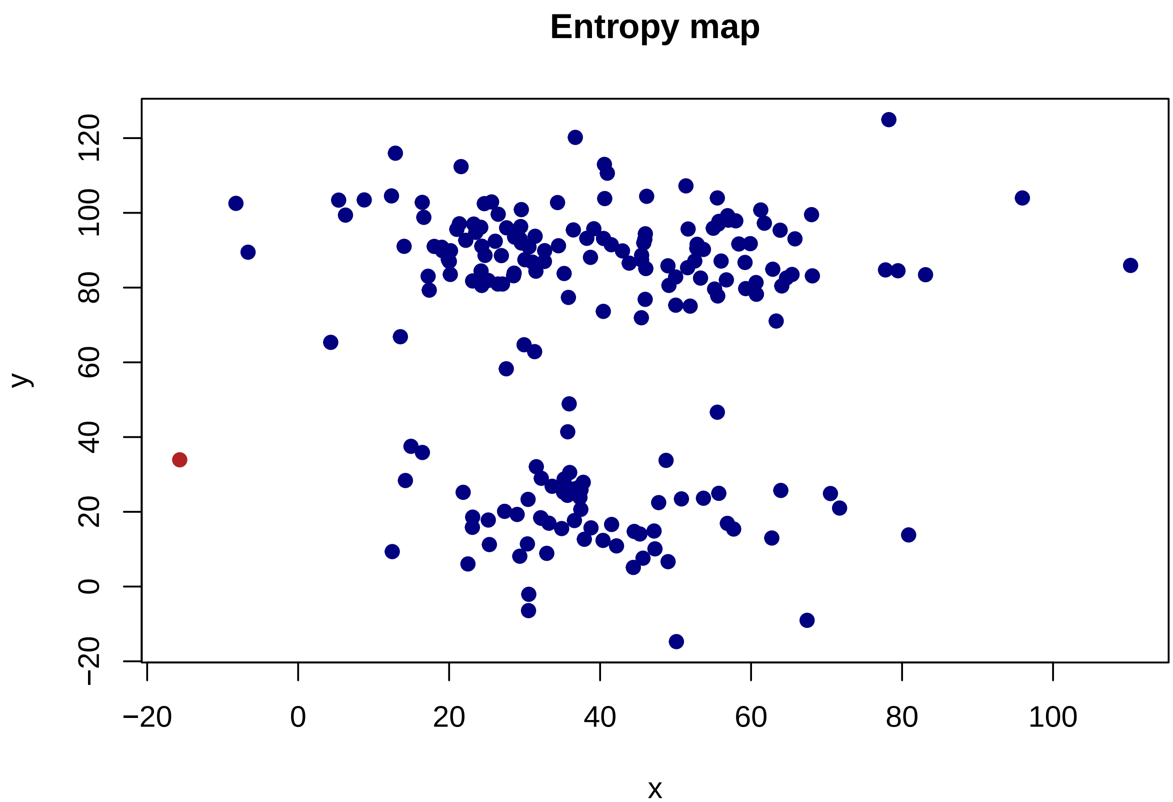
7.3 Base R plots

For environments without ggplot2:

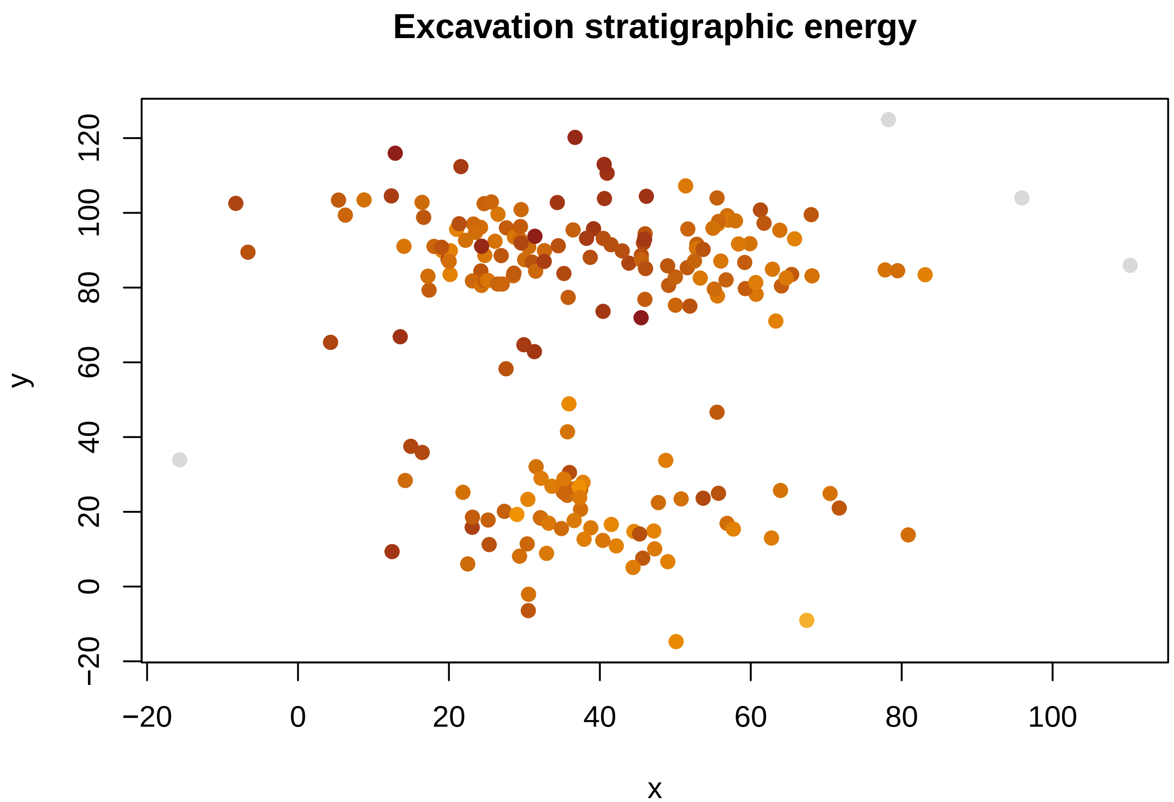
```r
plot_phasefield(fit)
```




```r
plot_entropy(fit)
```




```r
plot_energy(fit)
```




```
plot_sei_profile(fit)
```

7.4 GIS export

```
library(sf)

# Point layer with phase assignments
sf_pts <- as_sf_phase(fit, crs = 32632L, dims = "XYZ")
st_write(sf_pts, "results.gpkg", layer = "phases")

# High-SEI links as lines
sf_links <- as_sf_links(fit, quantile_threshold = 0.9, crs = 32632L)
st_write(sf_links, "results.gpkg", layer = "sei_links")
```

8 Interpretive Report

Generate a plain-language report for inclusion in excavation documentation:

```
report_sef(fit, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with **K = 3 phases** was fitted to a dataset of **200 fi
#>
#> The Palimpsest Dissolution Index (PDI) is **1.000**, indicating **excellent phase separation**. Mean
```

```
#>
#> Model diagnostics: LogLik = -1177.7, BIC = 2572.6.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (67 finds, 33.5%)
#>
#> - **Dating**: average 1148 CE (range: 1052 to 1246)
#> - **Material classes**: bone (17), ceramic (24), lithic (14), metal (12)
#> - **Mean entropy**: 0.0000; **Mean energy**: 1.938
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (66 finds, 33.0%)
#>
#> - **Dating**: average 1303 CE (range: 1211 to 1393)
#> - **Material classes**: bone (10), ceramic (22), lithic (20), metal (14)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.008
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (67 finds, 33.5%)
#>
#> - **Dating**: average 1001 CE (range: 894 to 1122)
#> - **Material classes**: bone (13), ceramic (21), lithic (24), metal (9)
#> - **Mean entropy**: 0.0000; **Mean energy**: 2.080
#> - **Contexts**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **11 finds** (5.5%) have an intrusion probability above 0.5, flagging them as potentially displaced
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | F024 | SU_1 | 0.663 |
#> | F112 | SU_9 | 0.625 |
#> | F062 | SU_3 | 0.569 |
#> | F145 | SU_1 | 0.560 |
#> | F013 | SU_1 | 0.556 |
#> | F120 | SU_2 | 0.547 |
#> | F012 | SU_3 | 0.531 |
#> | F090 | SU_9 | 0.530 |
#> | F048 | SU_2 | 0.519 |
#> | F080 | SU_2 | 0.506 |
#>
#> ## Stratigraphic Unit Purity
#>
#> Of **9 stratigraphic units**: **0 pure** (>90%), **0 mixed** (70-90%), **9 palimpsest** (<70%).
#>
#> Units classified as palimpsest require special attention, as material from multiple depositional epi
```

```
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
#> *Report generated by palimpsestr 0.19.0*
```

```
report_sef(fit, lang = "it")
#> # Rapporto Analisi SEF
#>
#> ## Panoramica
#>
#> Il modello Stratigraphic Entanglement Field con **K = 3 fasi** e' stato applicato a un dataset di **
#>
#> Il Palimpsest Dissolution Index (PDI) e' **1.000**, indicando una separazione **eccellente** delle f
#>
#> Diagnostiche: LogLik = -1177.7, BIC = 2572.6.
#>
#> ## Composizione delle Fasi
#>
#> ### Fase 1 (67 reperti, 33.5%)
#>
#> - **Datazione media**: 1148 d.C.
#> - **Classi materiali**: bone (17), ceramic (24), lithic (14), metal (12)
#> - **Entropia media**: 0.0000; **Energia media**: 1.938
#> - **US**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 2 (66 reperti, 33.0%)
#>
#> - **Datazione media**: 1303 d.C.
#> - **Classi materiali**: bone (10), ceramic (22), lithic (20), metal (14)
#> - **Entropia media**: 0.0000; **Energia media**: 2.008
#> - **US**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ### Fase 3 (67 reperti, 33.5%)
#>
#> - **Datazione media**: 1001 d.C.
#> - **Classi materiali**: bone (13), ceramic (21), lithic (24), metal (9)
#> - **Entropia media**: 0.0000; **Energia media**: 2.080
#> - **US**: SU_1, SU_2, SU_3, SU_4, SU_5, SU_6, ... (9 totali)
#>
#> ## Intrusioni
#>
#> **11 reperti** (5.5%) presentano probabilita' di intrusione superiore a 0.5.
#>
#> ## Purezza delle US
#>
#> Su **9 US**: **0 pure** (>90%), **0 miste** (70-90%), **9 palinsesto** (<70%).
#>
#> ---
#> *Rapporto generato da palimpsestr 0.19.0*
```

9 Database Import

Import data from SQLite or PostgreSQL:

```r
# SQLite
data <- read_db(
  con = DBI::dbConnect(RSQLite::SQLite(), "excavation.sqlite"),
  table = "finds"
)

# PostgreSQL (PyArchInit)
data <- read_db(
  con = DBI::dbConnect(RPostgres::Postgres(), dbname = "pyarchinit", host = "localhost"),
  table = "finds_table"
)
```

10 Phase Reordering

Mixture models suffer from *label switching*: phase numbers are arbitrary across runs. Use `reorder_phases()` to ensure phase 1 = deepest (oldest) and phase K = shallowest (most recent):

```r
fit <- reorder_phases(fit)
table(fit$phase)
#>
#>  1  2  3
#> 67 67 66
```

11 Bootstrap Confidence Intervals

Assess uncertainty on key metrics via bootstrap resampling:

```r
bs <- bootstrap_sef(fit, n_boot = 50, verbose = FALSE)
print(bs)
#>      statistic      estimate         lower         upper           se
#> 1          pdi  9.999945e-01  9.998558e-01  1.000000e+00 6.087534e-05
#> 2 mean_entropy  6.095726e-06  6.806146e-11  1.583963e-04 6.687840e-05
#> 3  mean_energy  2.008381e+00  1.880952e+00  2.022657e+00 4.105364e-02
#> 4       loglik -1.177707e+03 -1.190941e+03 -1.113264e+03 2.306197e+01
```

Returns a data.frame with columns `metric`, `mean`, `se`, `ci_lower`, `ci_upper` for PDI, mean entropy, mean energy, loglik, and optionally ARI (if `true_labels` is provided).

Parameters:

Parameter	Default	Description
`n_boot`	100	Number of bootstrap replicates
`conf`	0.95	Confidence level for intervals
`true_labels`	NULL	Known true phases for ARI confidence interval
`verbose`	TRUE	Print progress messages

12 Cross-Validation

Evaluate model fit via k-fold cross-validation:

```r
cv <- cv_sef(moderate, k_values = 3, n_folds = 3, seed = 42)
print(cv)
#>   k fold train_loglik test_loglik train_pdi
#> 1 3    1    -777.3906   -412.3507 0.9999997
#> 2 3    2    -761.4142   -439.0358 1.0000000
#> 3 3    3    -792.4940   -386.4890 0.9999999
```

Use with multiple K values to find the best:

```r
cv <- cv_sef(data, k_values = 2:5, n_folds = 5, seed = 42)
print(cv)  # shows mean_test_loglik per k
```

13 Data-Driven Weight Optimisation

Instead of manually tuning SEI weights, use `optimize_weights()` to find the best configuration via grid search:

```r
opt <- optimize_weights(data, k = 3, n_folds = 3, seed = 42)
print(opt$best_weights)

# Re-fit with optimal weights
fit <- fit_sef(data, k = 3, weights = opt$best_weights)
```

14 Sparse SEI for Large Datasets

For datasets with n > 1000, the full $n \times n$ SEI matrix may be too large. Use `sei_sparse()` which automatically applies a distance threshold:

```r
# Automatic sparse mode
sei <- sei_sparse(data, max_dist = 5.0)

# Or use sei_matrix() directly with max_dist
sei <- sei_matrix(data, max_dist = 5.0)
```

Pairs beyond `max_dist` Euclidean distance receive SEI = 0, dramatically reducing memory and computation time.

15 Complete Workflow Example

```r
library(palimpsestr)

# 1. Load data
data <- read.csv("site_data.csv")

# 2. Select optimal K
ck <- compare_k(data, k_values = 2:6, context = "context",
                tafonomy = "taf_score", n_init = 3)
gg_compare_k(ck)
best_k <- ck$k[which.min(ck$bic)]

# 3. Fit with optimal K and multiple inits
fit <- fit_sef(data, k = best_k, context = "context",
               tafonomy = "taf_score", n_init = 5, seed = 42)
```

```r
# 4. Inspect results
print(fit)
gg_phasefield(fit)
gg_entropy(fit)
gg_convergence(fit)
gg_phase_profile(fit)

# 5. Intrusion detection
gg_intrusions(fit, top_n = 10)

# 6. Harris validation
validate_phases_harris(fit)

# 7. Export
export_results(fit, dir = "results/")
report_sef(fit, lang = "en")

# 8. GIS
sf_pts <- as_sf_phase(fit, crs = 32632L)
sf::st_write(sf_pts, "results.gpkg", layer = "phases")
```

16 Real-Data Example: Villa Romana

The `villa_romana` dataset is derived from the multi-period Roman villa of Poggio Gramignano (Lugnano in Teverina, Italy): 615 inventoried finds across 54 stratigraphic units and 17 material classes, spanning the Pre-Roman to Late Antique periods, with excavator-assigned taphonomic scores. Note that coordinates and dating are recorded at stratigraphic-unit level (see `recommend_setup()`), so the model resolves phases at the unit and macro-event level.

16.1 Load and inspect

```r
data(villa_romana)
cat("Finds:", nrow(villa_romana), "| Contexts:", length(unique(villa_romana$context)),
    "| Classes:", length(unique(villa_romana$class)), "\n")
#> Finds: 615 | Contexts: 54 | Classes: 17
cat("Date range:", min(villa_romana$date_min), "to", max(villa_romana$date_max), "\n")
#> Date range: -650 to 550
table(villa_romana$class)
#>
#>            Campionature        Intonaco Dipinto Materiale Da Costruzione
#>                       5                       1                       38
#>       Materiale Lapideo       Pietre Dure-Gemme    Reperti Antropologici
#>                       1                       1                        1
#>  Reperti Archeozoologici       Reperto Anforaceo         Reperto Ceramico
#>                       7                      76                      417
#>      Reperto Faunistico         Reperto Lapideo        Reperto Metallico
#>                       1                       6                       19
#>     Reperto Numismatico          Reperto Vitreo  Resti Di Pavimentazioni
#>                      10                      25                        1
#>     Scarti Di Produzione                  Stucco
#>                       5                       1
```

16.2 Fit with taphonomic weighting

```
fit_vr <- fit_sef(villa_romana, k = 4,
                  tafonomy = "taf_score",
                  context  = "context",
                  n_init   = 5, seed = 42)
print(fit_vr)
#> <sef_fit>
#> Observations: 615
#> Estimated phases: 4
#> Mean entropy: 0.000
#> Mean local SEI: 1175.378
#> Mean energy: 0.882
#> LogLik: 3536.812
#> BIC: -6386.510
#> ICL: -6386.524
#> PDI: 1.000
#> Converged: yes
#> Class model: multinomial
#> Initialisations: 5
```

16.3 Diagnostics

```
cat("PDI:", pdi(fit_vr), "\n")
#> PDI: 0.9999916
sef_summary(fit_vr)
#> $n
#> [1] 615
#>
#> $k
#> [1] 4
#>
#> $pdi
#> [1] 0.9999916
#>
#> $mean_entropy
#> [1] 1.159321e-05
#>
#> $mean_energy
#> [1] 0.8821964
#>
#> $loglik
#> [1] 3536.812
#>
#> $bic
#> [1] -6386.51
#>
#> $phase_count
#>
#>   1   2   3   4
#> 436  48 121  10
```

16.4 Validation (on simulated data)

Validation with ARI and confusion matrix requires known true phases. We demonstrate with `demo_moderate`:

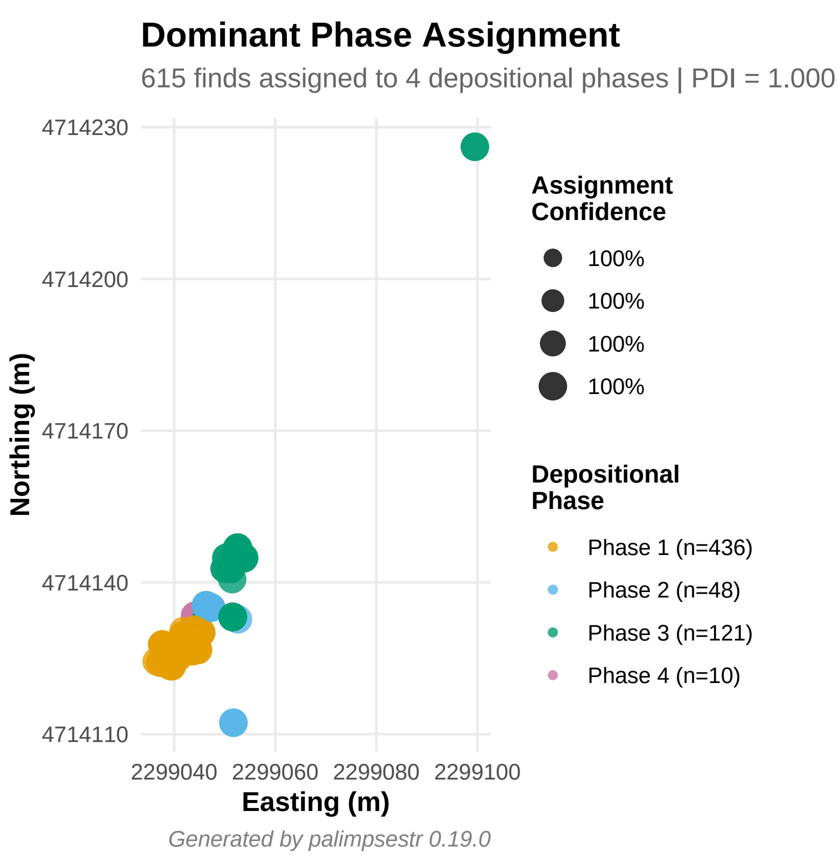
```
data(demo_moderate)
fit_demo <- fit_sef(demo_moderate, k = 3, seed = 42)
adjusted_rand_index(fit_demo, demo_moderate$true_phase)
#> [1] 1
confusion_matrix(fit_demo, demo_moderate$true_phase)
#>          true
#> estimated  2  1  3
#>         1 67  0  0
#>         2  0 67  0
#>         3  0  0 66
```

16.5 Intrusion detection

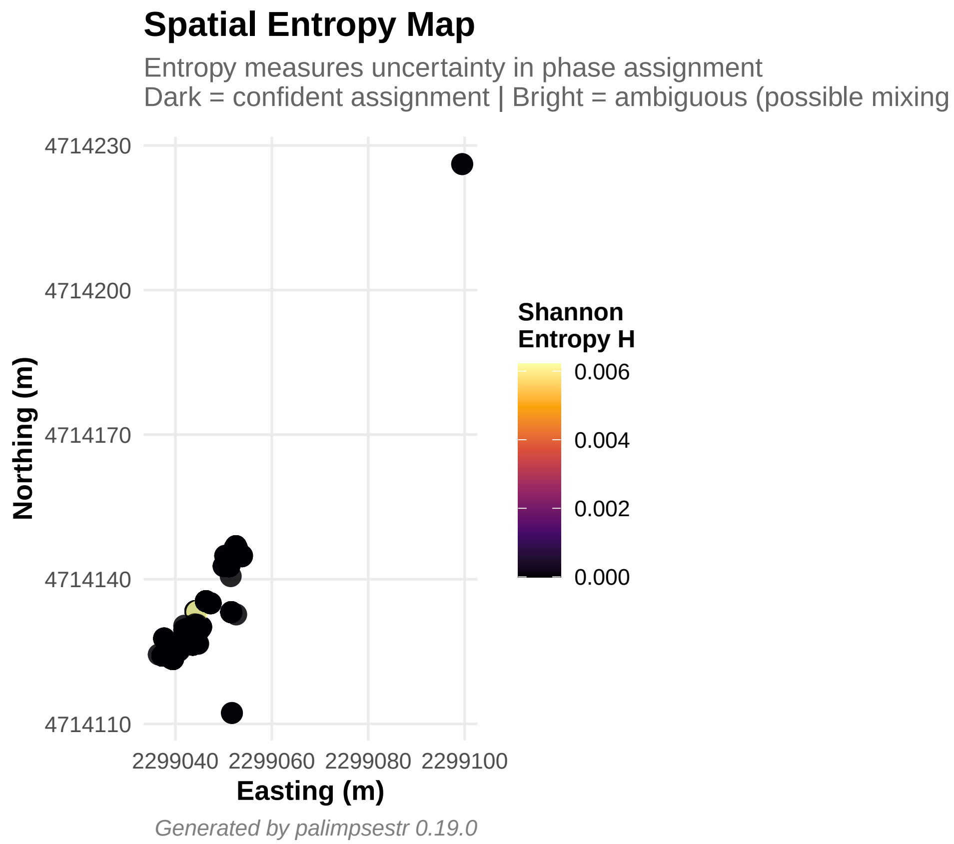
```
di_vr <- detect_intrusions(fit_vr)
suspects_vr <- di_vr[di_vr$intrusion_prob > 0.5, ]
cat("Suspected intrusions:", nrow(suspects_vr), "out of", nrow(villa_romana), "\n")
#> Suspected intrusions: 3 out of 615
```

16.6 Visualisation

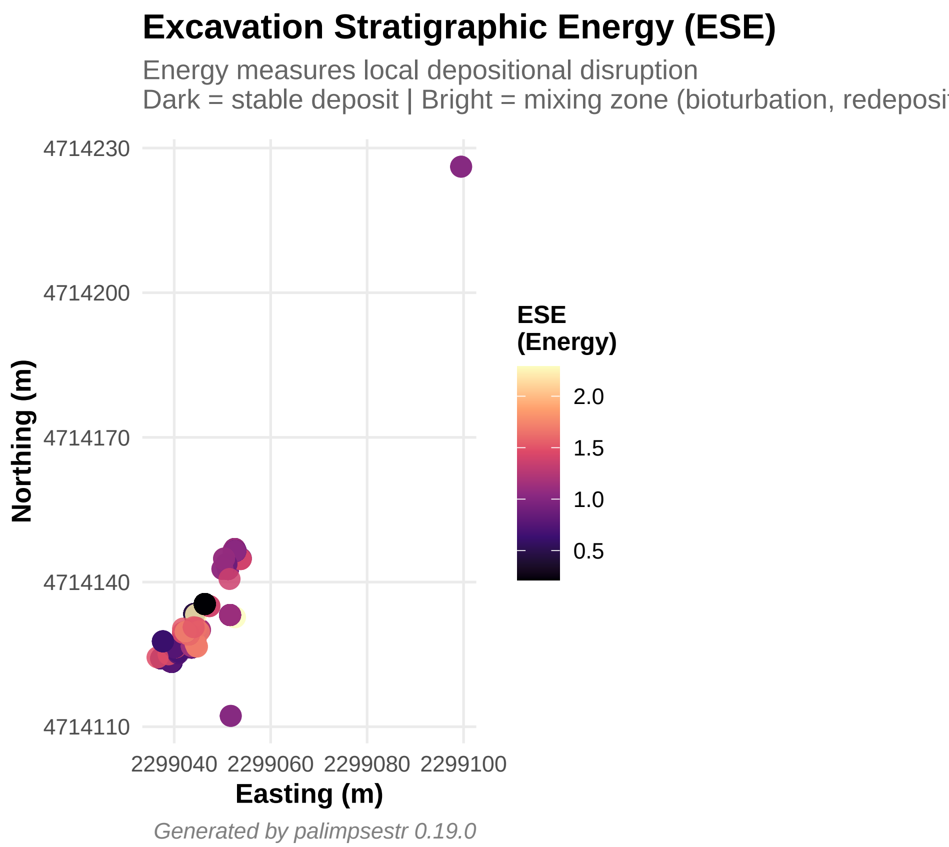
```
gg_phasefield(fit_vr)
```



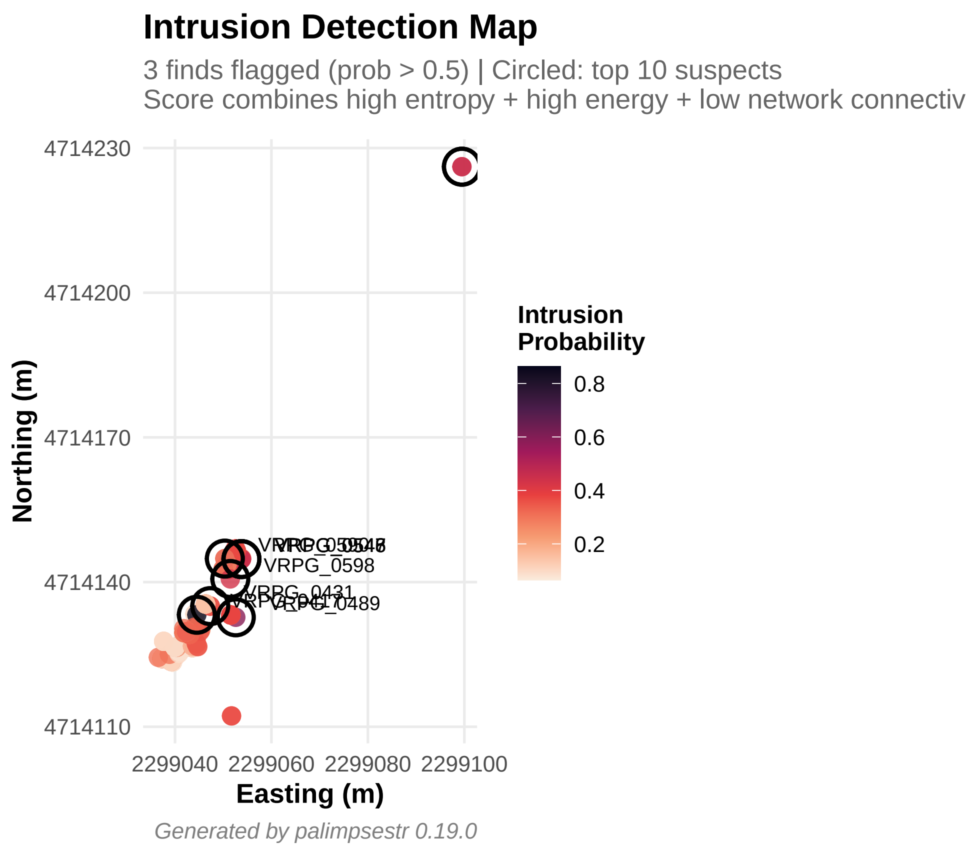
```r
gg_entropy(fit_vr)
```



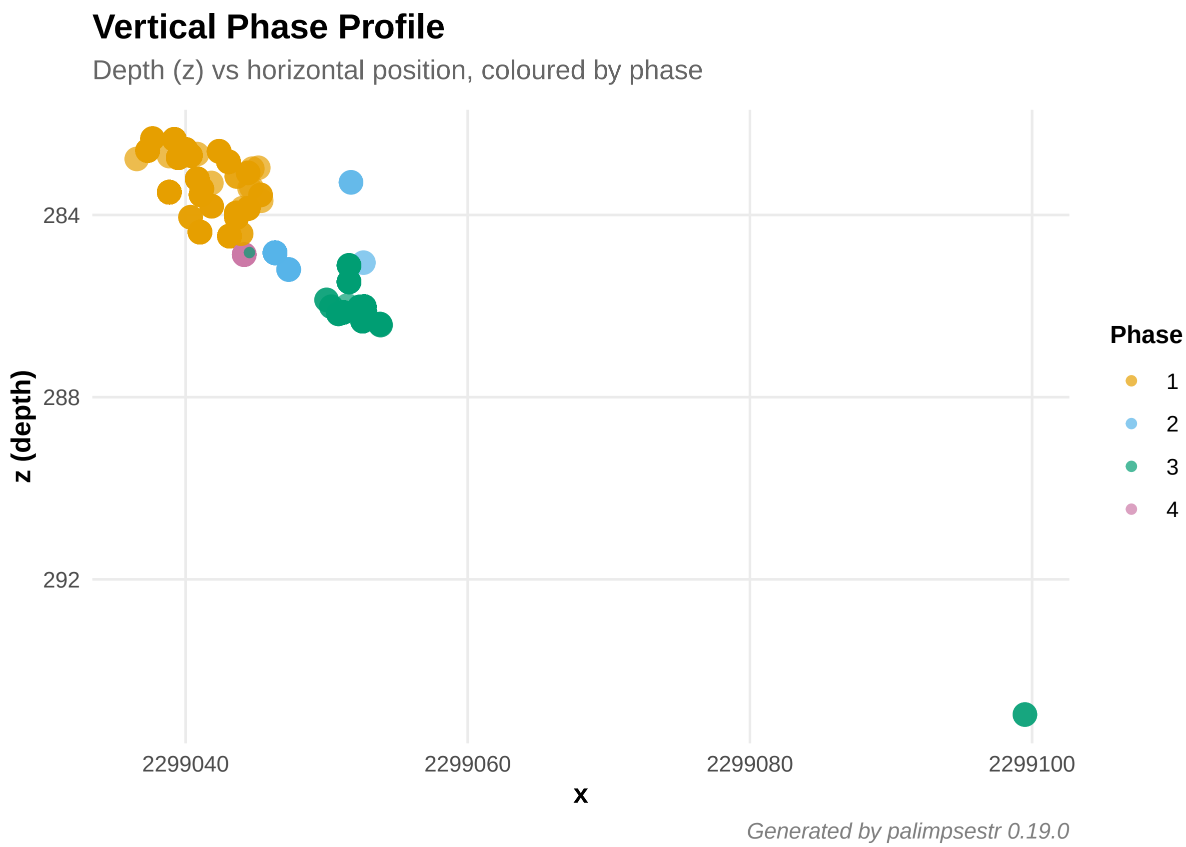

```r
gg_energy(fit_vr)
```




```r
gg_intrusions(fit_vr, top_n = 10)
```




```r
gg_phase_profile(fit_vr)
```



16.7 Per-US summary

```
us_summary_table(fit_vr)
#>    context n_finds dominant_phase purity mean_entropy mean_energy
#> 1   US_102      13              1      1 1.348904e-12   0.9116826
#> 2   US_104      93              1      1 8.006380e-18   0.7710822
#> 3   US_105      19              1      1 3.229626e-20   0.6686666
#> 4   US_108       3              1      1 3.517656e-14   0.8569123
#> 5   US_111      25              1      1 3.385322e-24   0.9672565
#> 6   US_114      22              1      1 6.073610e-15   0.8245524
#> 7   US_116       6              1      1 1.609169e-15   1.0170747
#> 8   US_117      27              1      1 7.201187e-23   0.9095915
#> 9   US_118      16              1      1 4.617782e-11   0.9341534
#> 10  US_133       1              1      1 3.567652e-21   1.4766197
#> 11  US_134      41              1      1 1.393958e-16   0.7658666
#> 12  US_136      33              1      1 9.487312e-10   0.9885305
#> 13  US_137      15              1      1 2.139375e-10   1.1168077
#> 14  US_144      18              1      1 3.009715e-10   1.1194293
#> 15  US_148       1              1      1 3.903770e-19   0.7056760
#> 16  US_149      15              1      1 1.766526e-19   0.6630704
#> 17  US_150      48              1      1 1.401461e-20   0.8866959
#> 18  US_151      10              4      1 2.189355e-10   0.5242270
#> 19  US_157       3              1      1 1.546477e-09   1.1795692
#> 20  US_158       1              1      1 6.542247e-10   1.1824122
```

```
#> 21  US_161       4            1       1 1.085694e-13   1.4024354
#> 22  US_166       2            1       1 5.154336e-11   1.3664971
#> 23  US_173       1            3       1 6.215833e-03   2.2285077
#> 24  US_180       1            1       1 4.676603e-25   1.4738468
#> 25  US_183       5            1       1 1.238610e-24   0.6177584
#> 26  US_189       2            1       1 4.967867e-14   1.3148516
#> 27  US_197       3            1       1 6.187898e-13   1.6754428
#> 28  US_201       1            1       1 2.153991e-14   1.5198388
#> 29  US_204       4            2       1 8.614016e-11   1.6094968
#> 30  US_209       1            1       1 2.767693e-11   1.4576088
#> 31   US_21       2            2       1 4.006853e-58   1.0000000
#> 32  US_210      41            2       1 0.000000e+00   0.2834099
#> 33  US_211       1            1       1 1.431351e-12   1.6650720
#> 34  US_215       1            1       1 3.888527e-15   1.6986328
#> 35  US_220       1            1       1 6.717062e-14   1.6964680
#> 36  US_258       8            1       1 5.487815e-15   1.5910054
#> 37  US_276       1            2       1 3.671074e-19   2.2891247
#> 38  US_287      25            3       1 1.166596e-07   0.9416951
#> 39  US_296      10            3       1 4.292317e-10   0.9686296
#> 40  US_297      17            3       1 1.371370e-09   0.8483098
#> 41  US_299       1            3       1 2.312728e-11   1.0080910
#> 42  US_302       2            3       1 0.000000e+00   1.0000000
#> 43  US_303       1            3       1 2.291997e-11   1.0066694
#> 44  US_304       3            3       1 3.588579e-12   1.3775978
#> 45  US_306       5            3       1 4.949377e-09   1.0256186
#> 46  US_307      30            3       1 3.959752e-09   0.9129552
#> 47  US_308       4            3       1 1.228439e-07   1.0316612
#> 48  US_315       2            3       1 9.861311e-06   1.1318639
#> 49  US_316       2            3       1 8.074350e-07   1.1944883
#> 50  US_319       6            3       1 1.059890e-04   0.9671551
#> 51  US_324       1            3       1 5.366125e-09   1.3699999
#> 52   US_72       5            1       1 9.967366e-17   1.4994895
#> 53   US_76      11            3       1 2.300811e-05   1.0012723
#> 54   US_89       1            1       1 6.263975e-11   1.5259711
#>    mean_local_sei n_intrusions
#> 1      1475.67927            0
#> 2      1454.47240            0
#> 3      1462.31198            0
#> 4      1489.49232            0
#> 5      1296.11532            0
#> 6      1440.29144            0
#> 7      1337.52512            0
#> 8      1339.40955            0
#> 9      1366.51078            0
#> 10     1092.23363            0
#> 11     1459.61396            0
#> 12     1376.46698            0
#> 13     1198.39339            0
#> 14     1236.64246            0
#> 15     1481.23740            0
#> 16     1440.03425            0
#> 17     1407.46372            0
#> 18     1273.79366            0
```

```
#> 19      1428.85433            0
#> 20      1076.40240            0
#> 21      1060.04979            0
#> 22      1027.26673            0
#> 23       606.90952            1
#> 24      1066.93451            0
#> 25      1362.36933            0
#> 26      1270.42593            0
#> 27      1041.68052            0
#> 28      1102.20707            0
#> 29       710.52499            1
#> 30      1091.58828            0
#> 31       592.93607            0
#> 32       854.27737            0
#> 33      1044.09577            0
#> 34      1034.54278            0
#> 35       998.15146            0
#> 36      1061.88830            0
#> 37       368.35027            1
#> 38       559.45797            0
#> 39       604.88564            0
#> 40       654.84219            0
#> 41       781.72219            0
#> 42        60.16121            0
#> 43       781.46404            0
#> 44       383.27231            0
#> 45       497.67760            0
#> 46       579.06540            0
#> 47       523.70113            0
#> 48       642.66694            0
#> 49       615.68993            0
#> 50       720.36507            0
#> 51       413.37717            0
#> 52      1204.92622            0
#> 53       618.10970            0
#> 54      1010.33897            0
```

16.8 Interpretive report

```
report_sef(fit_vr, lang = "en")
#> # SEF Analysis Report
#>
#> ## Overview
#>
#> The Stratigraphic Entanglement Field model with **K = 4 phases** was fitted to a dataset of **615 fi
#>
#> The Palimpsest Dissolution Index (PDI) is **1.000**, indicating **excellent phase separation**. Mean
#>
#> Model diagnostics: LogLik = 3536.8, BIC = -6386.5.
#>
#> ## Phase Composition
#>
#> ### Phase 1 (436 finds, 70.9%)
```

```
#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Materiale Da Costruzione (20), Reperto Anforaceo (62), Reperto Ceramico (319
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.912
#> - **Contexts**: US_102, US_104, US_105, US_108, US_111, US_114, ... (33 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 2 (48 finds, 7.8%)
#>
#> - **Dating**: average 374 BCE (range: -650 to 550)
#> - **Material classes**: Materiale Da Costruzione (3), Reperti Archeozoologici (1), Reperto Anforaceo
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.466
#> - **Contexts**: US_204, US_21, US_210, US_276
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 3 (121 finds, 19.7%)
#>
#> - **Dating**: average 372 CE (range: 180 to 425)
#> - **Material classes**: Campionature (5), Intonaco Dipinto (1), Materiale Da Costruzione (15), Mater
#> - **Mean entropy**: 0.0001; **Mean energy**: 0.971
#> - **Contexts**: US_173, US_287, US_296, US_297, US_299, US_302, ... (16 total)
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ### Phase 4 (10 finds, 1.6%)
#>
#> - **Dating**: average 438 CE (range: 425 to 450)
#> - **Material classes**: Reperto Anforaceo (1), Reperto Ceramico (9)
#> - **Mean entropy**: 0.0000; **Mean energy**: 0.524
#> - **Contexts**: US_151
#> - **Assessment**: Entropy is very low, suggesting this phase is internally coherent and well-defined
#>
#> ## Intrusion Detection
#>
#> **3 finds** (0.5%) have an intrusion probability above 0.5, flagging them as potentially displaced o
#>
#> Top suspects:
#>
#> | ID | Context | Intrusion Prob. |
#> |---|---|---|
#> | VRPG_0417 | US_173 | 0.864 |
#> | VRPG_0489 | US_276 | 0.593 |
#> | VRPG_0431 | US_204 | 0.578 |
#>
#> ## Stratigraphic Unit Purity
#>
#> Of **54 stratigraphic units**: **54 pure** (>90%), **0 mixed** (70-90%), **0 palimpsest** (<70%).
#>
#>
#> ## Recommendations
#>
#> The deposit shows good phase separation. The model can be used with confidence for phase attribution
#>
#> ---
```

```
#> *Report generated by palimpsestr 0.19.0*
```

17 Function Reference

17.1 Core

Function	Description
`fit_sef()`	Fit the SEF model
`compare_k()`	Compare multiple K values
`archaeo_sim()`	Simulate archaeological data
`reorder_phases()`	Relabel phases by mean depth (deepest = phase 1)

17.2 Diagnostics

Function	Description
`pdi()`	Palimpsest Dissolution Index
`sef_summary()`	Compact model summary
`as_phase_table()`	Phase probability table
`predict_phase()`	Alias for `as_phase_table()`
`phase_diagnostic_table()`	Full diagnostic table
`detect_intrusions()`	Intrusion detection

17.3 Validation

Function	Description
`adjusted_rand_index()`	Adjusted Rand Index
`confusion_matrix()`	Confusion matrix (auto-reordered)

17.4 Export

Function	Description
`us_summary_table()`	Per-US summary statistics
`phase_transition_matrix()`	Vertical transition counts
`export_results()`	Export all results to CSV

17.5 Harris Matrix

Function	Description
`harris_from_contexts()`	Auto-generate penalty from depths
`read_harris()`	Import Harris Matrix from CSV
`validate_phases_harris()`	Check phase–stratigraphy consistency

17.6 Robustness & Sensitivity

Function	Description
bootstrap_sef()	Bootstrap confidence intervals on PDI, entropy, ARI
cv_sef()	K-fold cross-validation for model selection
optimize_weights()	Data-driven SEI weight optimisation via grid search

17.7 SEI / ESE

Function	Description
sei_matrix()	Stratigraphic Entanglement Index matrix
sei_sparse()	Sparse SEI for large datasets (n > 1000)
local_sei()	Row sums of SEI matrix
ese()	Excavation Stratigraphic Energy

17.8 Plots (ggplot2)

Function	Description
gg_phasefield()	Phase assignment map
gg_entropy()	Entropy map
gg_energy()	ESE energy map
gg_intrusions()	Intrusion detection map
gg_compare_k()	K selection diagnostics
gg_convergence()	EM convergence trace
gg_phase_profile()	Vertical depth profile
gg_confusion()	Confusion matrix heatmap
gg_cv()	Cross-validation diagnostic plot
gg_bootstrap()	Bootstrap confidence interval plot
gg_weights()	Weight sensitivity heatmap
gg_map()	Overlay on excavation plan
as_plotly()	Convert any gg_* to interactive

17.9 Plots (base R)

Function	Description
plot_phasefield()	Phase map
plot_entropy()	Entropy map
plot_energy()	Energy map
plot_sei_profile()	SEI profile

17.10 GIS

Function	Description
as_sf_phase()	Export as sf point layer
as_sf_links()	Export SEI links as sf lines
load_geometries()	Load excavation plan geometries

17.11 Other

Function	Description
`report_sef()`	Interpretive report (en/it)
`read_db()`	Import from SQLite/PostgreSQL